

# Regression

## Assignment Questions

### 1. What is Simple Linear Regression?

ANS. Simple Linear Regression is a statistical method used to find the relationship between **one independent variable (X)** and **one dependent variable (Y)** by fitting a straight line to the data. It helps us predict the value of Y based on X.

The equation is  $Y = mX + c$ , where **m** is the slope and **c** is the intercept.

### 2. What are the key assumptions of Simple Linear Regression?

ANS. **Linearity:** The relationship between X and Y should be approximately linear.

**Independence:** Observations should be independent of each other.

**Homoscedasticity:** The variance of errors should be constant for all values of X.

**Normality of errors:** Residuals (errors) should follow a normal distribution.

**Feature relevance:** The independent variable (X) should have some meaningful relationship with Y — it shouldn't be completely unrelated.

### 3. What does the coefficient m represent in the equation $Y=mX+c$ ?

ANS. The coefficient m is called the slope of the line.

It represents the change in Y for a one-unit change in X.

- If  $m > 0$ , Y increases as X increases.
- If  $m < 0$ , Y decreases as X increases.

In simple words: **m tells us how steep the line is and the direction of the relationship.**

### 4. What does the intercept c represent in the equation $Y=mX+c$ ?

ANS. The intercept **c** is the point where the line **crosses the Y-axis**.

- It represents the **value of Y when X = 0**.
- In other words, it's the starting value of Y before any change in X.

Example: If you're predicting salary based on years of experience, **c** would be the expected salary when someone has **0 years of experience**.

### 5. How do we calculate the slope m in Simple Linear Regression?

ANS. The slope **m** is calculated as:

$$m = \frac{\sum (X_i - \bar{X}) * (Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

Where:

- **X<sub>i</sub>** = each value of X

- $Y_i$  = each value of Y
- $\bar{X}$  = mean of X
- $\bar{Y}$  = mean of Y

It tells us **how much Y changes for a one-unit change in X**.

This formula comes from the **least squares method**, which finds the line that **minimises the total squared differences** between predicted and actual Y values

#### 6. What is the purpose of the least squares method in Simple Linear Regression?

ANS. The **least squares method** is used to find the **best-fitting line** for the data.

- It works by **minimising the sum of the squared differences** between the actual values (Y) and the predicted values ( $\hat{Y}$ ) from the line.
- Squaring ensures that **both positive and negative differences** contribute positively.
- In short: it gives the line that **fits the data as closely as possible**.

#### 7. How is the coefficient of determination ( $R^2$ ) interpreted in Simple Linear Regression?

ANS. The **coefficient of determination ( $R^2$ )** measures how well the regression line **explains the variation in Y**.

- It ranges from **0 to 1**.
- **$R^2 = 0$** : The model does **not explain any variation** in Y.
- **$R^2 = 1$** : The model **explains all the variation** in Y.
- In simple terms:  **$R^2$  tells us the percentage of Y's variation that is explained by X**.

#### 8. What is Multiple Linear Regression?

ANS. Multiple Linear Regression is an extension of Simple Linear Regression where we use **two or more independent variables ( $X_1, X_2, \dots, X_n$ )** to predict a dependent variable (Y).

- The equation is:

$$Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + \dots + b_n \cdot X_n$$

- **$b_0$**  = intercept,  **$b_1, b_2, \dots, b_n$**  = coefficients for each feature.
- It helps us understand the **combined effect of multiple factors** on Y and make predictions.

#### 9. What is the main difference between Simple and Multiple Linear Regression?

ANS. Simple Linear Regression: Uses only one independent variable (X) to predict Y.

- **Multiple Linear Regression**: Uses **two or more independent variables ( $X_1, X_2, \dots$ )** to predict Y.

In short: **Simple = 1 feature, Multiple = multiple features**.

### 10. What are the key assumptions of Multiple Linear Regression?

ANS. Multiple Linear Regression has assumptions similar to Simple Linear Regression, plus a few additional ones:

1. **Linearity:** The relationship between each independent variable and Y should be linear.
2. **Independence:** Observations should be independent of each other.
3. **Homoscedasticity:** The variance of residuals should be constant across all levels of X.
4. **Normality of errors:** Residuals should be normally distributed.
5. **No multicollinearity:** Independent variables should **not be highly correlated** with each other.
6. **No major outliers:** Outliers should not strongly influence the model.

### 11. What is heteroscedasticity, and how does it affect the results of a Multiple Linear Regression model?

ANS. **Heteroscedasticity** occurs when the **variance of errors (residuals) is not constant** across all levels of the independent variables.

**Effects on the model:**

- The **estimates of coefficients remain unbiased**, but
- The **standard errors become unreliable**, which can lead to **wrong conclusions** in hypothesis tests (like t-tests) and confidence intervals.

**In simple terms:** It makes the model's predictions **less trustworthy** for some values of X.

### 12. How can you improve a Multiple Linear Regression model with high multicollinearity?

ANS. **Multicollinearity** happens when two or more independent variables are **highly correlated**. This makes it hard to know each variable's true effect on Y.

**Ways to fix or reduce it:**

1. **Remove one of the correlated variables** — keep the more important one.
2. **Combine variables** — create a single feature from correlated ones.
3. **Use regularization techniques like Ridge or Lasso regression.**
4. **Increase sample size** if possible (helps reduce variance).

### 13. What are some common techniques for transforming categorical variables for use in regression models?

ANS. Regression models need **numbers**, so categorical variables must be converted. Common techniques:

1. **One-Hot Encoding:**
  - Create a new column for each category (1 if present, 0 otherwise).
  - Example: Color = Red, Blue, Green → three columns: Red, Blue, Green.

## 2. Label Encoding:

- Assign a unique number to each category.
- Example: Red = 1, Blue = 2, Green = 3.
- Use carefully — model may assume **order matters**.

## 3. Binary Encoding / Dummy Variables:

- Similar to one-hot but reduces number of columns for high-cardinality features.

### 14. What is the role of interaction terms in Multiple Linear Regression?

ANS. **Interaction terms** show how two or more independent variables **together affect the dependent variable (Y)**, beyond their individual effects.

- Example: Predicting salary based on **experience (X1)** and **education level (X2)**.
  - Alone: X1 and X2 each affect salary.
  - Interaction: Maybe the effect of experience is **stronger for higher education levels**.
- In regression, we add a term like:

$$Y = b_0 + b_1 \cdot X_1 + b_2 \cdot X_2 + b_3 \cdot (X_1 \cdot X_2)$$

- **b3** measures the combined effect.

### 15. How can the interpretation of intercept differ between Simple and Multiple Linear Regression?

ANS. Simple Linear Regression:

- Intercept **c** is the value of Y when the single X = 0.

- **Multiple Linear Regression:**

- Intercept **b0** is the value of Y when **all independent variables (X1, X2, ...) = 0**.
- Sometimes this may not make real-world sense if 0 is **outside the data range**.

In short: The concept is the same, but in multiple regression, it depends on **all features being zero**, which may or may not be meaningful.

### 16. What is the significance of the slope in regression analysis, and how does it affect predictions?

ANS. The **slope(s)** (coefficient(s) of X) tells **how much the dependent variable Y changes** when the independent variable X increases by 1 unit, keeping other variables constant (in multiple regression).

- **Positive slope:** Y increases as X increases.
- **Negative slope:** Y decreases as X increases.
- The slope **directly affects predictions**, because it determines the **rate and direction of change** in Y based on X.

- In short: **Slope shows the strength and direction of the relationship between X and Y.**

**17. How does the intercept in a regression model provide context for the relationship between variables?**

ANS. The **intercept** (c or  $b_0$ ) gives the **baseline value of Y** when all independent variables are zero.

- It provides **context** because it tells us where the regression line **starts** on the Y-axis.
- Even if  $X = 0$  isn't meaningful in real life, the intercept **helps the model position the line correctly** to capture the relationship between X and Y.

In short: **Intercept sets the starting point of the relationship**, giving context for how Y changes as X changes.

**18. What are the limitations of using  $R^2$  as a sole measure of model performance?**

ANS.  **$R^2$  only measures fit:** It shows how much of Y's variation is explained by X, but **doesn't tell if the model is appropriate.**

- **Doesn't indicate bias or overfitting:** A high  $R^2$  can happen with irrelevant features or overfitting.
- **Insensitive to added variables:** In multiple regression,  $R^2$  **always increases** when you add more variables, even if they're useless.
- **No information about causality:** A high  $R^2$  doesn't mean X causes Y.

In short:  **$R^2$  is helpful but not enough alone; check residuals, adjusted  $R^2$ , and other metrics too.**

**19. How would you interpret a large standard error for a regression coefficient?**

ANS. A **large standard error** means the estimated coefficient is **less precise.**

- It indicates that **the coefficient could vary a lot** if we used a different sample of data.
- In practical terms: **we are less confident** about the effect of that variable on Y.
- This can lead to **wider confidence intervals** and **less significant p-values** in hypothesis testing.

In short: **Large standard error → low confidence in the coefficient estimate.**

**20. How can heteroscedasticity be identified in residual plots, and why is it important to address it?**

ANS. Identification:

- Plot **residuals vs predicted values.**
- **Homoscedasticity:** residuals are randomly scattered with **no clear pattern.**
- **Heteroscedasticity:** residuals show a **pattern**, like a funnel shape (spread increases or decreases with X).
- **Importance:**
  - Heteroscedasticity makes **standard errors unreliable**, leading to **wrong confidence intervals and p-values.**

- Correcting it improves **trustworthiness of predictions and tests**.

In short: **Look at residual plots; random scatter = good, pattern = problem.**

**21. What does it mean if a Multiple Linear Regression model has a high  $R^2$  but low adjusted  $R^2$ ?**

ANS. **High  $R^2$ :** The model explains a large portion of Y's variation.

- **Low adjusted  $R^2$ :** Some of the added independent variables **don't really improve the model**.
- **Reason:** Adjusted  $R^2$  **penalises unnecessary variables**. If you add irrelevant features,  $R^2$  increases but adjusted  $R^2$  may drop.

In short: **The model may be overfitting or have redundant features.**

**22. Why is it important to scale variables in Multiple Linear Regression?**

ANS. Scaling ensures all features are on a **similar range**, which is important because:

1. **Gradient-based optimization** (like in training with gradient descent) **converges faster**.
2. **Coefficients become comparable** — without scaling, a feature with large values may dominate the model.
3. Helps prevent **numerical instability** in calculations.

In short: **Scaling makes training more stable and coefficients easier to interpret.**

**23. What is polynomial regression?**

ANS. Polynomial regression is an extension of linear regression that models the **relationship between X and Y as a polynomial** (curved line) instead of a straight line.

- It allows the model to **fit non-linear patterns** in the data.
- Example:

$$Y = b_0 + b_1 * X + b_2 * X^2 + b_3 * X^3 + \dots$$

In short: **Polynomial regression captures non-linear relationships using powers of X.**

**24. How does polynomial regression differ from linear regression?**

ANS. **Linear regression:** Fits a **straight line** — Y changes linearly with X.

- **Polynomial regression:** Fits a **curved line** — Y changes non-linearly with X using powers of X ( $X^2$ ,  $X^3$ , etc.).
- **Use case:** When data shows **curved patterns**, linear regression may underfit, but polynomial regression can capture the curvature.

In short: **Linear = straight line, Polynomial = curve to fit non-linear data.**

**25. When is polynomial regression used?**

ANS. Polynomial regression is used when the relationship between the independent variable(s) and dependent variable is non-linear.

- Example cases:
  - Predicting **population growth** over time (curved trend).
  - Modeling **acceleration vs time** in physics.
  - Any dataset where a **straight line doesn't fit well**.

In short: **Use polynomial regression to capture curves in the data.**

## 26. What is the general equation for polynomial regression?

ANS. The general equation is:

$$Y = b_0 + b_1 * X + b_2 * X^2 + b_3 * X^3 + \dots + b_n * X^n$$

Where:

- **Y** = dependent variable
- **X** = independent variable
- **b<sub>0</sub>, b<sub>1</sub>, ... b<sub>n</sub>** = coefficients
- **n** = degree of the polynomial (controls the curve's flexibility)

In short: **Polynomial regression adds powers of X to model non-linear relationships.**

## 27. Can polynomial regression be applied to multiple variables?

ANS. Yes, polynomial regression can be extended to **multiple independent variables**.

- In this case, the model can include:
  - **Powers of each variable** ( $X_1^2$ ,  $X_2^3$ , etc.)
  - **Interaction terms** ( $X_1X_2$ ,  $X_1^2X_2$ , etc.)
- Example equation with two variables:

$$Y = b_0 + b_1 * X_1 + b_2 * X_2 + b_3 * X_1^2 + b_4 * X_1 * X_2 + b_5 * X_2^2$$

In short: **Polynomial regression can handle multiple variables and their interactions for non-linear patterns.**

## 28. What are the limitations of polynomial regression?

ANS.

1. **Overfitting:** High-degree polynomials can fit the training data too closely and **perform poorly on new data**.
2. **Extrapolation risk:** Predictions outside the data range can be **unreliable**.
3. **Complexity:** More terms = harder to interpret and compute.

4. **Sensitive to outliers:** Outliers can **distort the curve** significantly.

In short: **Polynomial regression is flexible but can overfit and be unstable.**

### 29. What methods can be used to evaluate model fit when selecting the degree of a polynomial?

ANS. To choose the best degree of a polynomial, you can use:

1. **Train-test split / Cross-validation:** Check **performance on unseen data** to avoid overfitting.
2. **Mean Squared Error (MSE) / Root MSE:** Lower values indicate better fit.
3. **Adjusted R<sup>2</sup>:** Accounts for the number of predictors; helps **penalise unnecessary complexity**.
4. **Visual inspection:** Plot the curve vs data to see if it captures the trend without overfitting.

In short: **Use metrics and visualisation to balance fit and simplicity.**

### 30. Why is visualisation important in polynomial regression?

ANS. Visualisation helps you **see the fit** of the polynomial curve to the data.

It helps identify:

1. **Underfitting:** Curve is too simple, misses trends.
2. **Overfitting:** Curve is too wiggly, follows noise.
3. **Outliers or anomalies:** Points that distort the curve.

It makes the model **interpretable** and easier to explain in interviews or presentations.

### 31. How is polynomial regression implemented in Python?

ANS. Use scikit-learn library:

**Steps:**

1. Import `PolynomialFeatures` and `LinearRegression`.
2. Transform your feature(s) to include polynomial terms.
3. Fit a linear regression model on the transformed features.

**Example:**

```
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
import numpy as np

# Sample data
```



```
X = np.array([1, 2, 3, 4, 5]).reshape(-1,1)
y = np.array([1, 4, 9, 16, 25])
```

```
# Transform features to polynomial (degree 2)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)
```

```
# Fit linear regression on transformed features
model = LinearRegression()
model.fit(X_poly, y)
```

```
# Predict
y_pred = model.predict(X_poly)
```

In short: **Polynomial regression = transform features to powers, then apply linear regression.**